Dilmini Warnakulasooriya

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Education

Georgia State University — USA
PhD student in Mathematics and Statistics — *Aug. 2023 – Present*
- GPA: 4.26/4.3
- Course highlights: Applied Mathematics, Systems Biology, Advanced Mathematical Biology, Advanced Matrix Analysis, Advanced Numerical Analysis, Applied Stochastic Processes

University of Moratuwa — Sri Lanka
Bachelor of Science (Hon) in Biomedical Engineering — *Feb. 2016 – Jan. 2020*
- GPA: 3.51/4.2 (Dean's List: Semester 6)
- Course Highlights: Digital Signal Processing, Fundamentals in Image Processing and Machine Vision, Genomic Signal Processing, Analysis of Physiology

Research Experience

Graduate Research Projects — 2023 - Present
- Title 1: *A compartmentalized mathematical model of failing mouse myocyte contraction* (2024/26)
- Title 2: *A compartmentalized mathematical model of hypertrophic mouse ventricular myocytes* (2023/24)
- Collection of experimental data, model development, fitting model to the experimental data, analyze the behavior of the electrical activity in hypertrophic mouse ventricular myocytes through mathematical modeling
- Research Advisor: Dr. Vladimir Bondarenko

Undergraduate Research Project (Thesis) — 2019
- Title: *Cuffless non-invasive method for blood pressure estimation*
- Technical tools used: Python (TensorFlow, Numpy, Scipy), SPSS
- Research Advisors: Dr. Nuwan Dayananda, Mr. Thilina Ambagahawatta

Internship Research Project — 2018
- Title: *Stress classification method based on non-EEG based physiological signals*
- Technical tools used: Matlab, Empatica E4 wrist band
- Supervisors: Mr. Pradeep Pathirana and Ms. Ayanga Kalupahana

Work Experience

Graduate Research Assistant — Oct 2024 - Present
Department of Mathematics and Statistics, GSU — *USA*

Graduate Teaching Assistant — Aug. 2025 – Dec 2025
College of Arts and Sciences, GSU — *USA*
- MATH0999 - Support for College Algebra

Graduate Lab Assistant — Aug. 2023 – Dec 2024, Aug. 2026 – Present
College of Arts and Sciences, GSU — *USA*
- Assist students in the lab to complete homework and quizzes and conduct review sessions

Instructor — Dec. 2020 – Jul. 2023
Department of Mathematics, University of Moratuwa — *Sri Lanka*
- Led weekly tutorial sessions, Graded assignments and exam papers, Invigilated examinations
- Mentored three groups of first-year undergraduate students

Trainee Research Engineer — Jun. 2018 – Dec. 2018
Synergen Technology Labs (Pvt) Ltd. — *Sri Lanka*
- Research project: Stress classification method based on non-EEG based physiological signals

Awards and Achievements

Outstanding MBD Fellow Award — Georgia, USA
Outstanding performance from the Department of Mathematics and Statistics — *2025*

Best Graduate Lab Assistant Award — Georgia, USA
Department of Mathematics and Statistics — *2024 - 2025*

MBD Doctoral Fellowship — Georgia, USA
Molecular Basis of Disease Program — *2024 - Present*

Women in Engineering - 'Build a Robot' Robotics Competition — Sri Lanka
2nd Runners-up — *2017*

Mahapola Merit Scholarship — Sri Lanka
National Level Performance at GCE A/level — *2016 - 2020*

Publications

Warnakulasooriya, D. and **Bondarenko, V.E.** *"EAD Mechanisms in Hypertrophic Mouse Ventricular Myocytes: Insights from A Compartmentalized Mathematical Model"*, Bulletin of Mathematical Biology, Feb 2025

Warnakulasooriya, D. and **Bondarenko, V.E.** *"The Effects of Stimulation of the β_1- and/or β_2-adrenoceptors on Mouse Ventricular Myocyte Contraction: Modeling Studies"*, Molecular Pharmacology, Dec 2025 (Under review)

Conference Presentations

11th Workshop on Bioinformatics and Biostatistics (Poster) 2026
Modeling the effects of vilanterol on mouse ventricular myocyte contraction *USA*

2026 Brain and Behavior Retreat, GSU (Poster) 2026
Comparative analysis of β-adrenergic stimulation of myocyte contraction by different agonists *USA*

2026 Molecular Basis of Disease Retreat, GSU (Oral) 2026
Comparative analysis of the effects of different β-agonists on the mouse ventricular myocyte contraction *USA*

2026 Georgia Scientific Computing Symposium (Poster) 2026
Comparative analysis of β-adrenergic stimulation of myocyte contraction by different agonists *USA*

4th Graduate Research Conference, GSU (Oral) 2026
An investigation of the effects of β_1- and β_2-adrenoceptor stimulation on mouse ventricular myocyte contraction *USA*

AMS Fall Southeastern Meeting 2025 (Oral) 2025
Compartmentalized mathematical model of hypertrophic mouse ventricular myocyte with β_1- and β_2-adrenergic signaling systems *USA*

7th Carnegie Mellon Forum on Biomedical Engineering (Oral) 2025
Mathematical modeling of isoproterenol and zinterol effects on mouse ventricular myocyte contraction *USA*

2025 Science and Cyberinfrastructure for Discovery Conference (Oral/ Poster) 2025
Mathematical modeling of EADs in hypertrophic mouse ventricular myocytes *USA*

10th Workshop on Bioinformatics and Biostatistics (Poster) 2025
Modeling the effects of isoproterenol and zinterol on mouse ventricular myocyte contraction *USA*

2025 Brain and Behavior Retreat, GSU (Poster) 2025
Mathematical modeling of early afterdepolarizations in mouse ventricular myocytes *USA*

2025 Molecular Basis of Disease Retreat, GSU (Oral) 2025
Compartmentalized mathematical model of hypertrophic mouse ventricular myocyte with β_1- and β_2-adrenergic signaling systems *USA*

Atlanta SIAM Student Conference (Poster) 2025
Mechanism of EADs in hypertrophic mouse ventricular myocytes *USA*

2025 Georgia Scientific Computing Symposium (Poster) 2025
Mechanism of EADs in hypertrophic mouse ventricular myocytes *USA*

3rd Graduate Research Conference, GSU (Oral) 2025
A compartmentalized mathematical model of hypertrophic mouse ventricular myocytes *USA*

6th Carnegie Mellon Forum on Biomedical Engineering (Oral) 2024
Mechanism of EADs in hypertrophic mouse ventricular myocytes: Insights from a mathematical model *USA*

2024 Brain and Behavior Retreat, GSU (Poster) 2024
Mathematical modeling of the β-adrenergic stimulation of hypertrophic mouse ventricular myocytes *USA*

2024 Georgia Scientific Computing Symposium (Poster) 2024
A compartmentalized mathematical model of hypertrophic mouse ventricular myocytes *USA*

Applied and Computational Mathematics Seminar, GSU (Oral) 2024
Mechanisms of EADs in hypertrophic mouse ventricular myocytes: Insights from a mathematical model *USA*

Memberships

Pi Mu Epsilon - GA Zeta chapter
Vice President (2026/Present), Student member *2025 - Present*

IEEE - EMBS Student Chapter
Student Member *2025 - Present*

Sri Lankan Student Association(SLaSA), Georgia State University
Member, Secretary (2024/25) *2023 - Present*

American Mathematical Society
Student Member *2023 - Present*

Skills

Computational Tools: R, Matlab, SPSS, Minitab, SciLab, ClustalX
Programming Languages: Python, C language, FORTRAN
Libraries: TensorFlow, OpenCV, pandas, NumPy, Scipy, Matplotlib